Ans. (b) : The historic Lahore session of Congress was held in December 1929 under the chairmanship of Pt. Jawaharlal Nehru. In this session, the attainment of complete Swaraj was approved as the goal.

14. The Ghadr (Ghadar) was a-

(a) revolutionary association of Indians with headquarters at San Francisco.
(b) nationalist organization operating from Singapore.
(c) militant organization with headquarters at Berlin.
(d) communist movement for India's freedom with head-quarters at Tashkent.

Ans. (a) Sohan Singh Bhakna founded the association called the 'Hindustan Association of the Pacific Coast' in San Francisco in 1913. This association took out a newspaper called 'Gadar' which also lended the name of this association as Gadar Party.

INDIAN GEOGRAPHY

• Physical Structure/Drainage System/ Human Geography

15. Which one of the following pairs of islands is separated from each other by the 'Ten Degree Channel' ?

(a) Andaman and Nicobar.
(b) Nicobar and Sumatra.
(c) Maldives and Lakshadweep.
(d) Sumatra and Java.

Ans. (a): The entire group of Islands is divided into two broad categories - the Andaman in the north and the Nicobar in the South. They are separated by a water body which is called the Ten Degree Channel.

16. If you travel through the Himalayas, you are likely to see which of the following plants naturally growing there ?

1. Oak 2. Rhododendron
3. Sandalwood

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1,2 and 3

Ans. (a): Sandalwood trees are found in the foothills of the Himalayas. The moist deciduous forests are more pronounced in the regions which record rainfall between 100-200 cm. These forests are found in the northeastern states along the foothills of Himalayas, eastern slopes of the Western Ghats and Odisha. Teak, sal, Shisham, Hurra, Mahua, Amla, Semul, Kusum and sandalwood etc. are the main species of these forests.

17. Which of the following have coral reefs ?

1. Andaman and Nicobar Islands
2. Gulf of Kachchh
3. Gulf of Mannar
4. Sunderbans

(a) 1, 2 and 3 only (b) 2 and 4 only
(c) 1 and 3 only (d) 1, 2, 3 and 4

Ans. (a): Coral reefs are a ridge of rock in the sea formed by the growth and deposit by coral. Major reef formations in India mainly occur in the Gulf of Mannar, Palk Strait, Gulf of Kutch, the Andaman and Nicobar Islands and the Lakshadweep.

18. Consider the following pairs :

Hills	Region
1. Cardamom Hills	Coromandel Coast
2. Kaimur Hills	Konkan Coast
3. Mahadeo Hills	Central India
4. Mikir Hills	North-East India

Which of the above pairs are correctly matched ?

(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 2 and 4

Ans. (c): The Cardamom Hills are southern hills of India and part of the Southwestern Ghats located in south east Kerala and southwest Tamil Nadu. They are not on the Coromandel coast.
Kaimur Range is the eastern portion of the Vindya Range extending from Madhya Pradesh to Bihar. They are not in Konkan Coast. The Mahadeo Hills are in Madhya Pradesh, Mikir Hills are in Assam, i.e., in North East India.

• Monsoon, Forest, Soils, Irrigation Projects, Agriculture

19. With reference to Neem tree, Consider the following statements :

1. Neem oil can be used as a pesticide to control the proliferation of some species of insects and mites.
2. Neem seeds are used in the manufacture of bio fuels and hospital detergents.
3. Neem oil has applications in pharmaceutical industries.

Which of the statements given above is /are correct ?

(a) 1 and 2 only (b) 3 only
(c) 1 and 3 (d) 1, 2 and 3

Ans. (c): Neem oil extracted by cold-pressing the seed kernels of Neem is highly effective against soft-bodied insects and mites. The presence of disulphide in Neem oil is a major contributor to its bioactivity.
The seeds of neem contain 30-40% oil. Biodiesel obtained from Neem oil which are mono alkyl esters produced using the 'Transesterification' process. The Neem seeds are also used as hospital detergents.
Neem oil used in the pharmaceutical industry contains various compounds that have medicinal, cosmetic and insecticidal properties.

• Minerals, Power Resources & Industry and Trade

20. With reference to technologies for solar power production, consider the following statements :

1. 'Photovoltaics' is a technology that generates electricity by direct conversion of light into electricity, while, 'Solar Thermal' is a technology that utilizes the Sun's rays to generate heat which is further used in electricity generation process.
2. 'Photovoltaics' generates Alternating Current (AC), while Solar Thermal generates Direct Current (DC).
3. India has manufacturing base for Solar Thermal technology, but not for 'Photovoltaics'.

Which of the statements given above is/ are correct ?

(a) 1 only (b) 2 and 3 only
(c) 1, 2 and 3 (d) None

Ans. (a): 'Photovoltaic's is a technology that generates electricity by direct conversion of light into electricity, while 'solar Thermal' is a technology that utilizes the Sun's rays to generate heat which is further used in the electricity generation process. Both Photovoltaic cells and solar thermal generate Direct Current (DC). India has a manufacturing base for both.

21. Which of the following are some important pollutants released by steel industry India ?

1. Oxides of sulphur 2. Oxides of nitrogen
3. Carbon monoxide 4. Carbon dioxide

Select the correct answer using the code given below :

(a) 1, 3 and 4 only (b) 2 and 3 only
(c) 1 and 4 only (d) 1,2,3 and 4

Ans. (d): In a steel furnace, coke reacts with iron ore, releasing iron and generating CO and CO_2 gases. Due to the use of coal, pollutants such as SO_x and NO_x are released; thus all the options are correct.

• Transport

22. Consider the following pairs.

National Highway		Cities connected
1. NH 4	:	Chennai and Hyderabad
2. NH 6	:	Mumbai and Kolkata
3. NH 15	:	Ahemdabad and Jodhpur

Which of the above pairs is /are correctly matched ?

(a) 1 and 2 only (b) 3 only
(c) 1,2 and 3 (d) None

Ans. (d): National Highway No. 4 connects Chennai to Thane (Mumbai). National Highway No. 6 connects Hazira (Gujrat) to Kolkata and National Highway No. 15 starts from Pathankot (Punjab) to Samakhiali (Gujrat).

WORLD GEOGRAPHY

• Evolution of Organism

23. Which of the following phenomena might have influenced the evolution of organisms?

1. Continental drift 2. Glacial cycles

Select the correct answer using the code given below :

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c): Both continental drift and Glacial cycles influenced the evolution of organisms. One of the most prominent examples of the effect of continental drift on the evolution of organisms is the unique primates found in Australia and South America. The final stages of the revolution of Genus Homo occur in the last there glacial cycles.

• Hydrosphere

24. Consider the following rivers :

1. Barak 2. Lohit 3. Subansiri

Which of the above flows /flow through Arunachal Pradesh?

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b) : The Barak is an important river in Manipur and Mizoram. The Barak river, the head- stream of the Meghna river in the hills in Manipur. Lohit river in Arunachal Pradesh in India. It is a tributary to the Brahmaputra river. The Subansiri River is a tributary of the Brahmaputra river in the states of Assam and Arunachal Pradesh.

• Climate Change

25. The seasonal reversal of winds is the typical characteristic of -

(a) Equatorial climate
(b) Mediterranean climate
(c) Monsoon climate
(d) All of the above climates

Ans. (c): A monsoon is a seasonal reversal in wind patterns over a region. A dramatic change in precipitation usually accompanies the seasonal wind shift. The best-known example of a monsoon occurs over the Indian subcontinent.
Monsoon refers to the seasonal reversal in the wind direction during a year.

26. The scientific view is that the increase in global temperature should not exceed 2°C above pre-industrial level, If the global temperature increases beyond 3°C above the pre-industrial level, what can be its possible impact/impacts on the world ?

1. Terrestrial biosphere tends toward a net carbon source.
2. Widespread coral mortality will occur.
3. All the global wetlands will permanently disappear.
4. Cultivation of cereals will not be possible anywhere in the world.

Select the correct answer using the code given below:

(a) 1 only (b) 1 and 2 only
(c) 2,3 and 4 (d) 1,2,3 and 4

Ans. (b): According to IPCC report, if there is an increase of 3ºC or more in temperature then the terrestrial biosphere will tend towards a net carbon source and if the world temperature rises more than 2ºC at pre-industrial level, then there will be widespread coral mortality around the world.

• Human Geography

27. Every year a month long ecologically important campaign/festival is held during which certain communities/ tribes plant saplings of fruit-bearing trees. Which of the following are such communities/tribes ?

(a) Bhutia and Lepcha
(b) Gond and Korku